MILITARY SPECIFICATION

ARMOR: GLASS REINFORCED PLASTIC LAMINATES

1. SCOPE

1.1 This specification covers Glass Reinforced Plastic Laminates for lightweight composite armor applications.

1.2 The plastic laminates shall be classified according to the following types:

Type I - Plastic laminates individually bonded and subsequently adhesively bonded to a ceramic component.
Type II - Plastic laminates molded and adhesively bonded to a ceramic component in a single operation.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-R-7575 - Resin, Polyester, Low Pressure Laminating
MIL-A-46165 - Armor: Woven Glass Roving Fabrics

STANDARDS

FEDERAL
Fed. Test Method Std. No. 406 - Plastics: Methods of Testing

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Insp ction by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity of as directed by the contracting officer.)

/FSC 9330/

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National Motor Freight Traffic Association, Inc., Agent:
National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:
Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Preproduction. Unless otherwise specified (see 6.2) a preproduction sample shall be required. The preproduction sample shall conform to all requirements in this specification.

3.2 Construction. The plastic laminate shall consist of plies of "woven roving" fabric impregnated and bonded together with a suitable thermosetting unsaturated polyester type resin.

3.2.1 Woven glass roving fabric reinforcement. The glass fabric reinforcement shall consist of "woven roving" fabric conforming to the requirements of MIL-A-46165.

3.2.2 Resin. The resin shall conform to the requirements of MIL-R-7575 Grade A, Glass O, Form A or B.

3.3 Resin content. The resin content plus any volatile or combustible finish (sizing) material shall be between 20 and 27 percent of the total laminate weight (see 4.8.1).

3.4 Specific gravity. The plastic laminate shall have a minimum specific gravity of 1.92 (see 4.8.2).

3.5 Temperature resistance. The plastic laminate shall show no evidence of delamination or degradation following exposure to a temperature range of minus 65 to plus 160 degrees F. (see 4.8.3).

3.6 Flammability. The plastic laminate shall be nonburning or self extinguishing (see 4.8.4).

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3.7 Ballistic requirements. The plastic laminates shall meet or exceed the ballistic requirement contained in the Classified Supplement of this specification (see 6.4).

3.8 Dimensions.

3.8.1 Lateral. The length and width of flat laminates shall conform to the dimensions and tolerances specified in the contract. The dimensions of contoured laminates shall conform to the drawings specified in the contract (see 6.2).

3.8.2 Thickness. Unless otherwise specified the thickness at any point more than one inch from an edge, shall not vary from the average thickness by more than plus and minus .010 inches. The average thickness, unless otherwise specified, shall be taken care of by the areal density and specific gravity requirements (see 6.2).

3.9 Areal density. The plastic laminate shall have an areal density as specified in the contract (see 6.2).

3.10 Identification marking.

3.10.1 Type I laminates. The plastic laminates shall be ink stamped or otherwise encoded with the areal density and manufacturers identification. The materials and procedures used in marking shall not impair the laminate properties.

3.10.2 Type II laminates. The plastic laminate may be identified as in 3.10.1 or may be identified with the ceramic component to which it is bonded if suitable encoding exists thereon to ensure proper identification.

3.11 Workmanship. The plastic laminate shall be uniform in thickness and free from wrinkles, blisters, cracks, crazing, fabric tears and chipped edges. The rovings shall be reasonably straight, evenly spaced and perpendicular warp-to-fill.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as

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otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be manufactured by the same methods proposed for the manufacture of subsequent production lots. The preproduction sample shall be subjected to all examinations and tests specified herein. If the preproduction sample is rejected, it will be returned to the contractor. Before submitting another preproduction sample, the contractor shall furnish the Government full particulars concerning previous rejection and the action taken to correct the defects. Subsequent quantities will not be considered for acceptance until the approval of the preproduction sample has been obtained.

4.3 Lot. A lot shall conform to the definition specified in MIL-STD-105.

4.4 Sampling.

4.4.1 Resin content. The inspection level shall be II of MIL-STD-105 and the Acceptable Quality Level shall be 2.5.

4.4.2 Shear strength. The inspection level shall be S-1 of MIL-STD-105 and the Acceptable Quality Level shall be 2.5.

4.4.3 Specific gravity. The inspection level shall be II of MIL-STD-105 and the Acceptable Quality Level shall be 2.5.

4.4.4 Temperature resistance. The inspection level shall be S-1 of MIL-STD-105 and the Acceptable Quality Level shall be 2.5.

4.4.5 Flammability. The inspection level shall be S-1 of MIL-STD-105 and the Acceptable Quality Level shall be 2.5.

4.4.6 Ballistic testing. Unless otherwise specified a minimum of four armor laminates will be ballistically tested for each lot. Ballistic test samples will be randomly selected from the laminates, with no failure permitted in meeting ballistic requirements.

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4.4.7 Dimensions. The inspection level shall be III of MIL-STD-105 and the Acceptable Quality Level shall be 4.0.

4.4.8 Identification marking. The inspection level shall be III of MIL-STD-105 and the Acceptable Quality Level shall be 4.0.

4.4.9. Areal density. The inspection level shall be III of MIL-STD-105 and the Acceptable Quality Level shall be 4.0.

4.5 Examination.

4.5.1 Visual. The sample laminates shall be visually examined for compliance with the workmanship requirements of 3.11.

4.5.2 Dimensions and areal density. The sample laminates shall be examined for compliance with the requirements of 3.8 and 3.9.

4.5.3 Preparation for delivery. Prior to shipment, examination shall be made for compliance with the requirements of section 5.

4.6 Test specimens.

4.6.1 Type I laminates. In case of type I laminates a release agent shall be utilized during manufacture such that bonding of the plastic laminate to the ceramic component does not occur nor does the releasing agent interfere with subsequent adhesive bonding to the ceramic component.

4.6.2 Contoured laminates. In the case of contoured laminates, type I or type II, specimens may be produced as flat panels of the same thickness as the plastic laminates and approximately 12 inches square. The test laminates shall be manufactured simultaneous with the lot represented, if possible, and shall utilize the same materials, manufacturing equipment and techniques.

4.7 Test conditions. Unless otherwise specified all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, a temperature of 60 to 90 degrees F, and relative humidity of 90 percent or less.

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vinyl chloride

4.8 Test methods.

4.8.1 Resin content. The resin content of the plastic laminate shall be determined in accordance with Federal Test Method Standard No. 406 Method 7061.

4.8.2 Specific gravity. The specific gravity of the plastic laminate shall be determined in accordance with Federal Test Method Standard No. 406 Method 5011.

4.8.3 Temperature resistance. The temperature resistance of the plastic laminate shall be determined by the following method. The test sample shall be heated in an oven set at 160 degrees for six hours. The test specimen shall be then allowed to cool to room temperature followed by cold exposure at minus 65 degrees F. for six hours. The specimen shall be allowed to warm to room temperature (60 to 90 degrees F.) and shall then be examined at the edges for evidence of delamination.

4.8.4 Flammability. The flammability of the plastic laminate shall be determined in accordance with Federal Test Method Standard No. 406 Method 2021.

4.8.5 Ballistic testing. The laminated armor samples shall be subjected to ballistic testing in accordance with the supplement to this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing. Unless otherwise specified packing shall be level C.

5.1.1 Level C. Packing shall be in accordance with commercial practice adequate to ensure acceptance and delivery by the carrier for the mode of transportation employed. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules as applicable.

5.2 Marking for shipment. In addition to any special marking specified in the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 The plastic laminate covered by this specification is intended for use as backing in opaque lightweight ceramic composite armor applications.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, type and date of this specification
(b) Destination of preproduction test sample, if required (see 3.1)
(c) Dimensions and tolerances required (see 3.8)
(d) Areal density required (see 3.9)
(e) Test conditions, if required (see 4.7)
(f) Packing instructions, if required (see 5.1)
(g) Additional marking (see 5.2).

6.3 The materials and procedures used for identification marking should be compatible with materials and techniques to be used in manufacture of the composite armor.

6.4 Conformance to the requirements of this specification may not assure that the plastic laminates will meet the specified ballistic requirement.

CUSTODIANS:
Army - MR

Review activities:
Army - ME, MU

User activities:
AT, GL

Preparing activity:
Army - MR

Project No. 9330-A562

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION
MIL-A-46166(MR), Armor: Glass Reinforced Plastic Laminates

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A
[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?
[ ] YES [ ] NO IF "YES" IN WHAT WAY?

4 REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.*)

SUBMITTED BY (*Printed or typed name and activity*) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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